Kedros Analytics

STAFFING WITH QUEUEING THEORY

How Many Staff on a Saturday?

Your till already records every customer arrival. Turn it into an hour-by-hour roster that cuts idle wages and walk-outs together.

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THE OPPORTUNITY

Three is too few at lunch and too many at breakfast

The roster pays for both mistakes at once: wages for idle hours, and customers who see the queue and walk out.

THE HABIT

“It’s always been three.”

“We tried four once and they stood around.”

All of it can be true at once. The roster is copied from last month and changes only when someone says lunchtime is unbearable.

A NUMBER PER HOUR

How many people the demand actually needs.

Queueing theory, the Erlang C formula call centres have used for a century, applied to the timestamps your till already records. Wages are usually the biggest cost an SME can actually control.

WHAT IT IS

From “it’s always been three” to a head-count for every hour

Published by Agner Krarup Erlang in 1917 for telephone exchanges, and used to staff call centres ever since.

1

Your till tickets

Timestamp, transaction type, how long it took and its value. Six months is enough to start.

2

Erlang C per hour

Arrivals and service time for every weekday and hour give the minimum staff for a stated promise.

3

A gap grid and a price

Current roster against required, with idle wages and lost customers in euros.

The hard floor: arrivals per hour × service time is how many staff are busy serving. Below it, the queue grows without limit.

Demo: a café, 26 weeks, 76,381 clean tickets (about 2,940 customers a week). Synthetic data.

WHAT YOU GAIN

Six ways it pays off for a small business

1 Arrivals per hour

Every ticket timestamp is an arrival. “Busy” becomes 64 an hour, give or take 13.

2 Measured service time

A takeaway takes two minutes, a return five. The mix sets the service rate.

3 A hard floor

Offered load is the minimum head-count. Below it, the queue never clears.

4 Waits and walk-outs

The chance of waiting, the share served in time and the average wait, per hour.

5 Staff for a promise

The smallest team that serves 80% within five minutes, for every hour slot.

6 A price on both mistakes

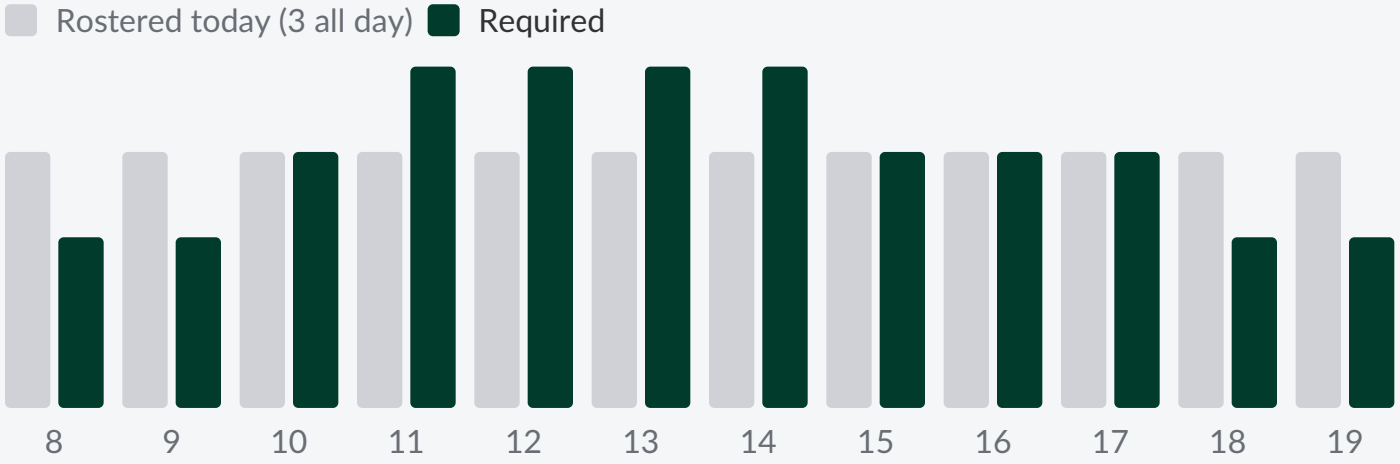
Idle wages plus lost customers, so rosters compare on one total cost.

Highlighted: the two that turn a head-count into a decision.

EXAMPLE RESULTS

Saturday has the right hours in the wrong places

Staff rostered today against staff required to serve 80% of customers within five minutes.



36 hours needed, 36 rostered

Same wage bill. Hiring or cutting would not fix it, only moving hours does.

4 idle hours, 4 queue hours

One too many at 8–10 and 18–20, one too few from 11 to 15.

Synthetic demo data (fictitious café). Bars are scaled to head-count.

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ONE HOUR UNDER THE MICROSCOPE

Saturday at noon: three people, a thirteen-minute wait

64.4 customers an hour, 157 seconds each: on average 2.82 people are busy serving at any moment.

SERVED WITHIN 5 MIN

37.4%

Against an 80% target.

Three people are 93.9% busy, and 88.7% of arriving customers have to wait at all.

AVERAGE WAIT

763 s

Nearly thirteen minutes.

The 13:00 slot is nearly as bad: 44.7% served in time, a 586-second wait.

CUSTOMERS LOST A WEEK

28.5

Worth €189.50 in contribution.

From this one slot, assuming customers give up after ten minutes.

With a fourth person, both noon slots serve 95–96% within five minutes and Saturday's walk-outs fall from 64 a week to 6.

Synthetic demo data. Wage €14 an hour, margin 65%, patience 10 minutes: all illustrative.

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THE STAFFING GAP

Three patterns most SME rosters share

Current roster minus required staff, per hour slot: red where short, amber where idle.

SATURDAY

36 vs 36

Right hours, wrong places.

Staff-hours rostered against needed. Idle at 8–10 and 18–20, short from 11 to 15.

WEEKDAY EVENINGS

~€750

Short every day, 17:00 to 19:00.

Lost contribution a week. The roster drops to two at 15:00, but the after-work peak needs three.

FRIDAY

27 vs 35

Rostered like a Tuesday.

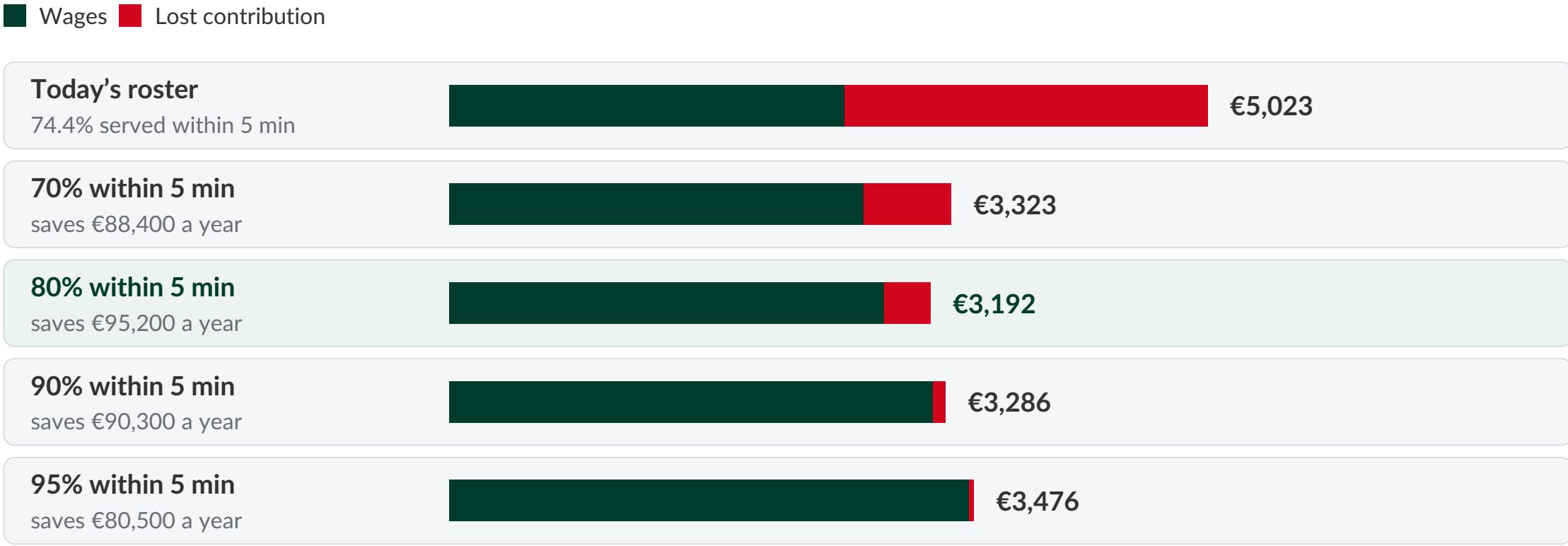
Staff-hours rostered against needed. At 17:00 and 18:00 two people cannot keep up at all.

Sunday is the same story in miniature: two people all day against a noon rate that needs three.

THE PRICE OF THE PROMISE

The cheapest promise to keep is the one in the middle

Weekly cost of each roster: wages plus lost contribution from customers who walk out.



Synthetic demo data. Each target is the share of customers served within five minutes.

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BUSINESS IMPACT

Same team, better shape, a lot more money

The demo café: 187 staff-hours a week rostered where 206 are needed.

€95,000

a year, for 19 more staff-hours a week. Served within five minutes goes from 74.4% to 92.7%.

€266 more in wages a week

Recovers about €2,100 a week in lost contribution.

361 walk-outs a week today

Invisible on the payroll. No one logs the customer who didn’t buy.

Illustrative: €14 an hour, 65% margin, 10 minutes’ patience. Use your own figures.

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GET STARTED

What to do this week

Five steps from a till export to a roster grid.

1

Export

Six months of tickets: id, timestamp, transaction type, duration and value.

2

Write it down

Your current roster as a grid: weekday × hour × people on shift.

3

Read the gap

Red is queues, amber is idle pay. Move hours from amber to red before hiring.

4

Pick the promise

Choose a target on the cost comparison, then build shifts around the grid.

5

Repeat

Re-run each quarter, or after any change to opening hours or menu.

No duration in your system? Time twenty of each transaction type with a stopwatch and use the averages.

KEEP IT HONEST

The limits worth keeping in mind

The grid is a well-founded starting point, not a law.

1 The model simplifies

Erlang C assumes random arrivals and patient customers. Real arrivals come in clumps, and real customers leave.

2 A grid is not a shift plan

Breaks, set-up, training and employment rules are missing. Building lawful, humane shifts is the next job.

3 Your money assumptions

Wage, margin and customer patience drive the euro figures. Use your own before quoting a saving.

4 Six months is one season

A café's December is not its June. Re-run when the season, hours or menu change.

HOW KEDROS ANALYTICS CAN HELP

Try it on your own data, with a free proof of concept

Business questions first, AI as a tool and not a hype machine, and results you can act on straight away.

1

Validate

Check your till or booking export for clock errors, open tickets and impossible values, so the model starts from data you can defend.

2

Model the demand

Estimate arrivals and service times per hour, and compare your roster with the grid your promise requires.

3

Roster and track

Move the hours that matter, then track waits, walk-outs and cost as seasons change.

Free PoC: tailored to your requirements and your company's analytics maturity.

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Ready to see how many people your shift needs?

Let’s spend thirty minutes on your till data. No commitment, no jargon, and we might even plan a free PoC.

Book a free call

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